



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Lab 4

Getting Started With Simulation Software Proteus

Contents

1	Introduction	2
2	Proteus	2
3	Getting Started with Proteus	2
3.1	Install Proteus	2
3.2	Create a File	2
3.3	Adding Basic Components to your Workspace	2
4	Binary Subtractor	4
5	Experiment 4: Implementation of Subtractor Circuits	5
5.1	Objectives	5
5.2	Tasks	5
6	Instructions:	6

1 Introduction

In the previous labs, we implemented various Boolean circuits using a trainer kit. From now on, all tasks will be performed individually using Proteus, a simulation software. A classroom assignment will be posted at the beginning of each lab session. You will have about 1 hour to complete the task and submit it with the proper file naming format. Each task will be evaluated based on a specific rubric.

Warning: If you are caught copying another classmate's lab task, you will receive a zero for that specific lab.

2 Proteus

Proteus is an electronic design automation (EDA) software developed by Labcenter Electronics. It combines circuit simulation, schematic capture, and PCB layout in a single platform. Widely used in education and industry, Proteus is especially known for simulating microcontroller-based designs, allowing users to test both hardware and firmware before building real prototypes.

3 Getting Started with Proteus

3.1 Install Proteus

You can install proteus from the [provided drive link](#) or any other source.

3.2 Create a File

1. Open Proteus.
2. Go to **New Project**, choose file name and path and click next.
3. Choose "Create a schematic from the selected template" and then choose "Default".
4. Then choose "Do not create a PCB layout" and click next.
5. Choose "No firmware project" and proceed.
6. Click "Finish".
7. If you get something similar to [Figure 1](#) then you have been able to create a file properly.

3.3 Adding Basic Components to your Workspace

1. Click on the **P** button on the left bar (next to devices).
2. Add the basic Input and Output components: LOGICTOGGLE for input and LOGICPROBE(BIG) for output. You can search for these components using keywords and add them by double-clicking on the desired item in the results on the right.

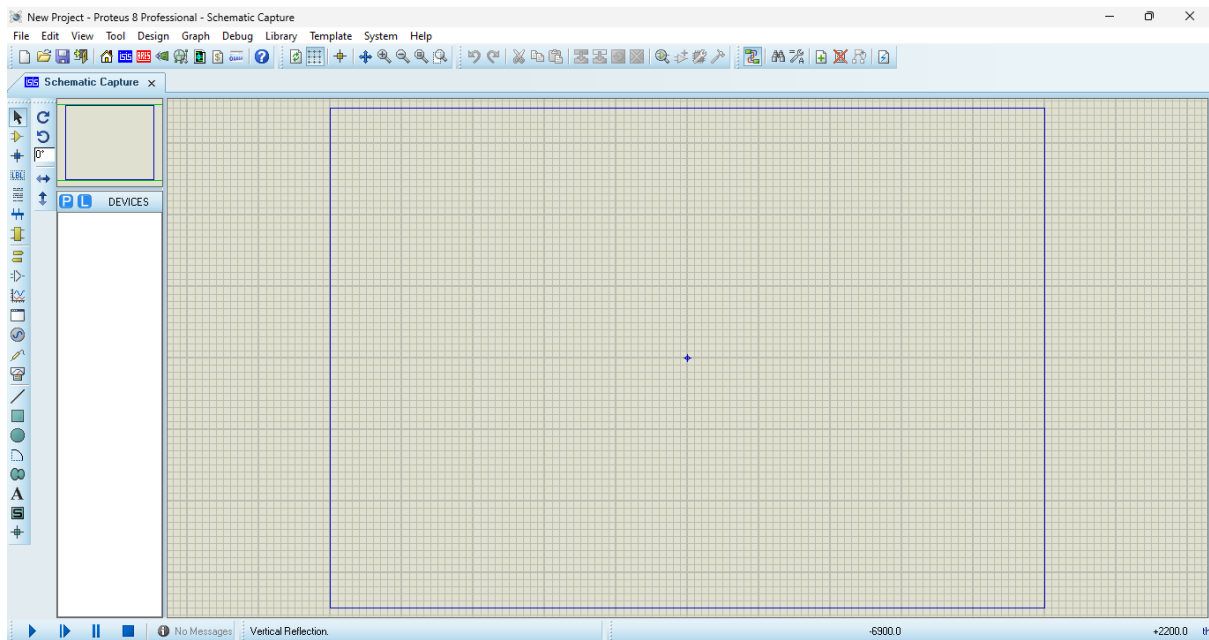


Figure 1: Empty Proteus File

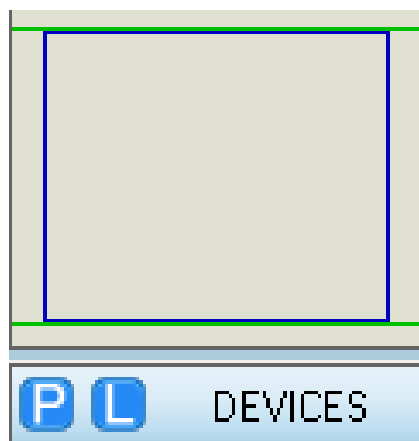


Figure 2: Pick Devices

3. To check a sample circuit, add another component, for example a X-OR gate in the similar way and build the following circuit.

To place the component in your workspace, choose a device from the DEVICES panel and click in your workplace. Connect the components using wire.

4. Press on the simulation icon and try out different combinations.

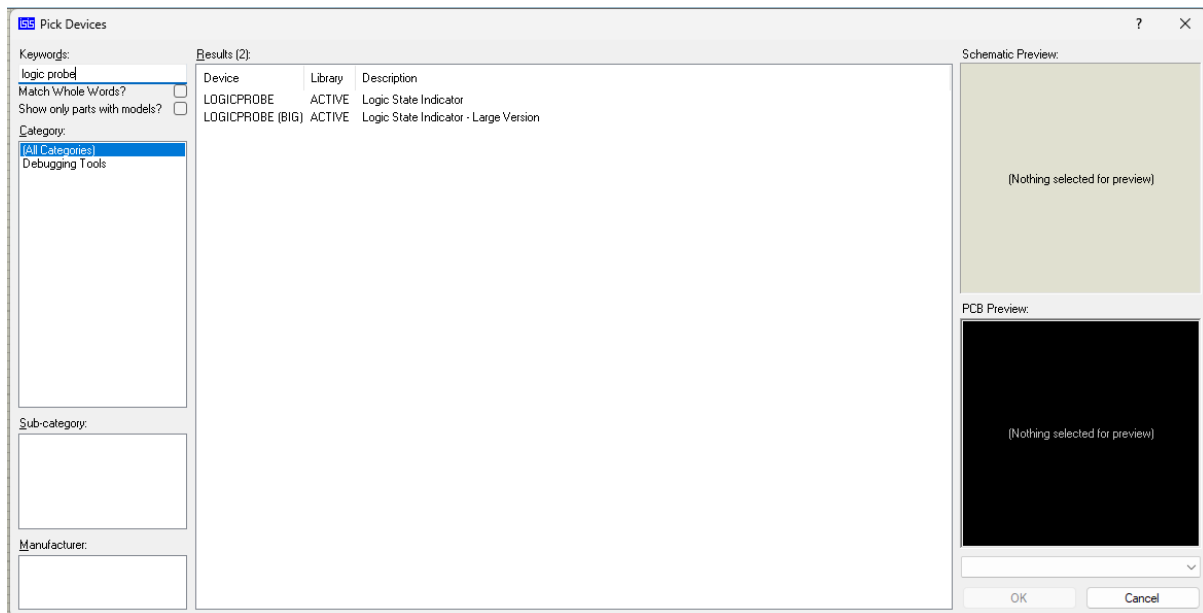


Figure 3: Adding Components

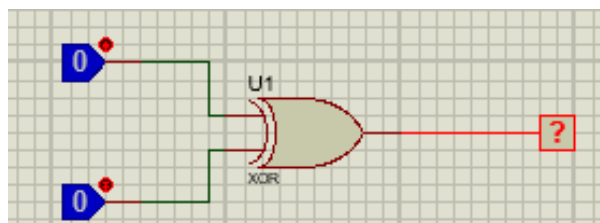


Figure 4: Basic X-OR Circuit

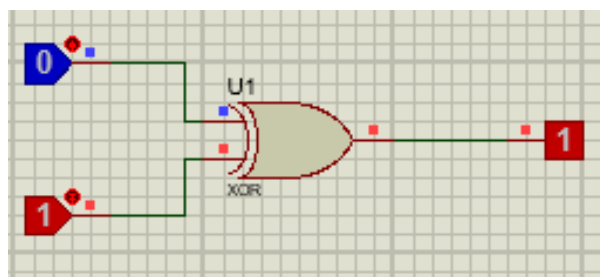


Figure 5: Simulation

4 Binary Subtractor

A binary subtractor is a digital logic circuit that performs subtraction of binary numbers. Just like an adder adds binary values, a subtractor computes the difference between two binary digits.

There are two main types:

1. **Half Subtractor:**

Subtracts two single bits A and B.

2. **Full Subtractor:**

Subtracts three bits: A, B and Borrow-in (B_{in}) from a previous stage.

5 Experiment 4: Implementation of Subtractor Circuits

5.1 Objectives

The objectives of this experiment are as follows:

1. To understand and implement binary Half Subtractor and Full Subtractor circuits.
2. To design and implement a 4-bit binary adder.

5.2 Tasks

Today's lab task are as follows -

1. Implement a Full Subtractor circuit using only basic gates (AND, OR, NOT)
2. Design a combinational circuit that compares two n -bit binary numbers A and B , where $n \geq 3$. The circuit should determine which number is larger and produce a single output bit G , **For Example**,

$$G = \begin{cases} 1, & \text{if } A > B, \\ 0, & \text{if } A \leq B. \end{cases}$$

You are **not allowed** to use a magnitude comparator IC or any predefined comparator block. The circuit must be implemented using only the combinational logic components and design techniques covered in class (e.g., logic gates, half adders, full adders, and related combinational circuits).

Your design should be scalable and work for any $n \geq 3$.

Bonus: Extend your design to generate three outputs indicating whether

$$A > B, \quad A = B, \quad A < B.$$

6 Instructions:

1. **File Naming Format:** ID_L04T01
2. You will be provided with an A4 sheet on which you must **write your ID at the top**. For Task 1 and Task 2, include the following:
 - (a) Truth Table
 - (b) Boolean expressions of the simplified circuit
 - (c) Simplification (if required) using K-map or cut and try method.
3. For Task 2, you must show the thought process behind the circuit design, providing appropriate justification.
4. You will get 1 hour to complete these tasks.
5. The rubrics for today's lab task can be found [here](#).